

CampusX Data Science Mentorship Program

Week 1: Basics of Python Programming

1. Session 1: Python Basics

- Short info about DSMP (3:38 – 6:57)
- About Python (7:00 – 24:30)
- Python Output/print function (24:30 – 38:37)
- Python Data Types (38:37 – 51:25)
- Python Variables (51:25 – 1:04:49)
- Python comments (1:04:49 – 1:09:09)
- Python Keywords and Identifiers (1:09:09 – 1:22:38)
- Python User Input (1:22:38 – 1:35:00)
- Python Type conversion (1:35:00 – 1:47:29)
- Python Literals (1:47:29 – 2:10:22)

2. Session 2: Python Operators + if-else + Loops

- Start of the session (00:00:00 – 00:09:02)
- Python Operators (00:09:02 – 00:43:00)
- Python if-else (00:43:00 – 01:14:50)
- Python Modules (01:14:50 – 01:24:48)
- Python While Loop (01:24:48 – 01:48:03)
- Python for loop (01:48:03 – 02:11:34)

3. Session 3: Python Strings

- Introduction (00:00:00 – 00:09:08)
- Solving Loop problems (00:00:08 – 00:47:10)
- Break, continue, pass statement in loops (00:47:10 – 01:06:42)
- Strings (01:06:42 – 1:14:15)
- String indexing (01:14:15 – 01:18:14)
- String slicing (01:18:14 – 01:27:06)
- Edit and delete a string (01:27:06 – 01:32:14)
- Operations on String (01:32:14 – 01:47:24)

- Common String functions (01:47:14 – 02:22:53)

4. Session on Time complexity

- Start of the Session (00:00:00 – 00:11:22)
- PPT presentation on Time Complexity (Efficiency in Programming and Orders of Growth) (00:11:22 – 01:12:30)
- Examples (01:12:30 – 01:42:00)

5. Week 1 Interview Questions

Week 2: Python Data Types

1. Session 4: Python Lists

- Introduction
- Array vs List
- How lists are stored in a memory
- Characteristics of Python List
- Code Example of Lists
 - Create and access a list*
 - append(), extend(), insert()*
 - Edit items in a list*
 - Deleting items from a list*
 - Arithmetic, membership and loop operations on a List*
 - Various List functions*
 - List comprehension*
 - 2 Ways to traverse a list*
 - Zip() function*
 - Python List can store any kind of objects*
- Disadvantages of Python list

2. Session 5: Tuples + Set + Dictionary

- **Tuple**
 - Create and access a tuple*
 - Can we edit and add items to a tuple?*

- iii. *Deletion*
- iv. *Operations on tuple*
- v. *Tuple functions*
- vi. *List vs tuple*
- vii. *Tuple unpacking*
- viii. *Zip () on tuple*

- **Set**

- i. *Create and access a set*
- ii. *Can we edit and add items to a set?*
- iii. *Deletion*
- iv. *Operations on set*
- v. *set functions*
- vi. *Frozen set (immutable set)*
- vii. *Set comprehension*

- **Dictionary**

- i. *Create dictionary*
- ii. *Accessing items*
- iii. *Add, remove, edit key-value pairs*
- iv. *Operations on dictionary*
- v. *Dictionary functions*
- vi. *Dictionary comprehension*
- vii. *Zip() on dictionary*
- viii. *Nested comprehension*

3. Session 6: Python Functions

- Create function
- Arguments and parameters
- args and kwargs
- How to access documentation of a function
- How functions are executed in a memory
- Variable scope
- Nested functions with examples
- Functions are first-class citizens
- Deletion of function
- Returning of function
- Advantages of functions
- Lambda functions

- Higher order functions
- map(), filter(), reduce()

4. Paid Session on Career QnA

5. Array Interview Questions

6. Week 2 Interview Questions

Week 3: Object Oriented Programming (OOP)

1. Session 7: OOP Part1

- What is OOP?
- What are classes and Objects?
- Banking application coding
- Methods vs Functions
- Class diagram
- Magic/Dunder methods
- What is the true benefit of constructor?
- Concept of 'self'
- Create Fraction Class
- `__str__`, `__add__`, `__sub__`, `__mul__`, `__truediv__`

2. Session 8: OOP Part2

- Revision of last session by solving problems
- How objects access attributes
- Attribute creation from outside of the class
- Reference Variables
- Mutability of Object
- Encapsulation
- Collection of objects
- Static variables and methods

3. Session 9: OOP Part3

- Class Relationship
- Aggregation and aggregation class diagram
- Inheritance and Inheritance class diagram
- Constructor example
- Method Overriding
- Super keyword
- Super constructor
- Practice questions on Inheritance

- Types of Inheritance (Single, Multilevel, Hierarchical, Multiple)
- Hybrid Inheritance
- Code example and diamond problem
- Polymorphism
- Method Overriding and Method Overloading
- Operator Overloading

4. Session on Abstraction

- What is Abstraction?
- Bank Example Hierarchy
- Abstract class
- Coding abstract class (BankApp Class)

5. Session on OOP Project

6. Week 3 Interview Questions

Week 4: Advanced Python:

1. Session 10: File Handling + Serialization & Deserialization

- How File I/O is done
- Writing to a new text file
- What is open()?
- append()
- Writing many lines
- Saving a file
- Reading a file -> read() and readline()
- Using context manager -> with()
- Reading big file in chunks
- Seek and tell
- Working with Binary file
- Serialization and Deserialization
- JSON module -> dump() and load()
- Serialization and Deserialization of tuple, nested dictionary and custom object
- Pickling
- Pickle vs JSON

2. Session 11: Exception Handling

- Syntax Error with Examples
- Exception with Examples

- Why we need to handle Exception?
- Exception Handling (Try-Except-Else-Finally)
- Handling Specific Error
- Raise Exception
- Create custom Exception

3. Session 12: Decorators and Namespaces

- Namespaces
- Scope and LEGB rule
- Hands-on local, enclosing, global and built-in scope
- Decorators with Examples

4. Session on Iterators

- What are iterators
- What are iterables
- How for loop works in Python?
- Making your own for loop
- Create your own range function
- Practical example to use iterator

5. Session on Generator

- What is a generator?
- Why to use Generator?
- Yield vs Return
- Generator Expression
- Practical Examples
- Benefits of generator

6. Session on Resume Building

7. Session on GUI Development using Python

- GUI development using tkinter

8. Week 4 Interview Questions

Week 5: Numpy

1. Session 13: Numpy Fundamentals

- Numpy Theory
- Numpy array
- Matrix in numpy
- Numpy array attributes
- Array operations

- Scalar and Vector operations
- Numpy array functions
 - Dot product*
 - Log, exp, mean, median, std, prod, min, max, trigo, variance, ceil, floor, slicing, iteration*
 - Reshaping*
 - Stacking and splitting*

2. Session 14: Advanced Numpy

- Numpy array vs Python List
- Advanced, Fancy and Boolean Indexing
- Broadcasting
- Mathematical operations in numpy
- Sigmoid in numpy
- Mean Squared Error in numpy
- Working with missing values
- Plotting graphs

3. Session 15: Numpy Tricks

- Various numpy functions like sort, append, concatenate, percentile, flip, Set functions, etc.

4. Session on Web Development using Flask

- What is Flask library
- Why to use Flask?
- Building login system and name entity recognition with API

Week 6: Pandas

1. Session 16: Pandas Series

- What is Pandas?
- Introduction to Pandas Series
- Series Methods
- Series Math Methods
- Series with Python functionalities
- Boolean Indexing on Series
- Plotting graphs on series

2. Session 17: Pandas DataFrame

- Introduction Pandas DataFrame
- Creating DataFrame and read_csv()

- DataFrame attributes and methods
- Dataframe Math Methods
- Selecting cols and rows from dataframe
- Filtering a Dataframe
- Adding new columns
- Dataframe function – astype()

3. Session 18: Important DataFrame Methods

- Various DataFrame Methods
- Sort, index, reset_index, isnull, dropna, fillna, drop_duplicates, value_counts, apply, etc.

4. Session on API Development using Flask

- What is API?
- Building API using Flask
- Hands-on project

5. Session on Numpy Interview Question

Week 7: Advanced Pandas

1. Session 19: GroupBy Object

- What is GroupBy?
- Applying builtin aggregation functions on groupby objects
- GroupBy Attributes and Methods
- Hands-on on IPL dataset

2. Session 20: Merging, Joining, Concatenating

- Pandas concat method
- Merge and join methods
- Practical implementations

3. Session on Streamlit

- Introduction to Streamlit
- Features of Streamlit
- Benefits of Streamlit
- Flask vs Streamlit
- Mini-project on Indian Startup Funding Dataset using Streamlit – Part 1

4. Session on Pandas Case Study (Indian Startup Funding)

- Data Analysis on Indian Startup Funding Dataset and display results on the Dashboard made by Streamlit – Part 2

5. Session on Git:

- What is Git?
- What is VCS/SCM?
- Why Git/VCS is needed?
- Types of VCS
- Advantages
- How Git works?
- Installing git
- Creating and Cloning repo
- add, commit, add ., gitignore
- seeing commits (log -> oneline)
- Creating versions of a software

6. Session on Git and GitHub:

- Nonlinear Development (Branching)
- Merging branches
- Undoing changes
- Working with a remote repo

Week 8: Advanced Pandas Continued

1) Session 21: MultiIndex Series and DataFrames

- About Multiindex objects
- Why to use Multiindex objects
- Stacking and unstacking
- Multiindex DataFrames
- Transpose Dataframes
- Swaplevel
- Long vs wide data
- Pandas-melt

2) Session 22: Vectorized String Operations | Datetime in Pandas

- Pivot table
- Agg functions
- Vectorized String operations
- Common functions
- Pandas Datetime

3) Session on Pandas Case Study – time Series analysis

4) Session on Pandas Case Study – Working with textual data

Week 9: Data Visualization

1) Session 23: Plotting Using Matplotlib

- Get started with Matplotlib
- Plotting simple functions, labels, legends, multiple plots
- About scatter plots
- Bar chart
- Histogram
- Pie chart
- Changing styles of plots

2) Session 24: Advanced Matplotlib

- Colored Scatterplot
- Plot size, annotations
- Subplots
- 3D plots
- Contour plots
- Heatmaps
- Pandas plot()

3) Session on Plotly (Express)

- About Plotly
- Disadvantages
- Introduction about Plotly Go, Plotly Express, Dash
- Hands-on Plotly

4) Session on Plotly Graph Objects (go)

5) Session on Plotly Dash

- Basic Introduction about Dash

6) Making a COVID-19 dashboard using Plotly and Dash

7) Deploying a Dash app on Heroku

8) Session on Project using Plotly

- Project using Indian Census Data with Geospatial indexing Dataset

Week 10: Data Visualization Continued

1) Session 25: Plotting Using Seaborn- part 1

- Why seaborn?
- Seaborn roadmap
- Main Classification plotting
- Relational plots
- Distribution plots
- KDE plot
- Matrix plot

2) Session 26: Plotting Using Seaborn- Part 2

- Categorical Plots
- Stripplot
- Figure level function – catplot
- Swarmplot
- Categorical Distribution Plots
- Boxplot
- Violinplot
- Categorical Estimate Plot – for central tendency
- Barplot
- Pointplot
- Countplot
- Faceting
- Doubt – error bar issue
- Regression Plots
- Regplot
- Lmplot
- Residual Plot
- FacetGrid
- Pairplot and Pairgrid
- JointGrid and Jointplot
- Utility Function in Seaborn – load_dataset
- Blog idea

3) Session on Open-Source Software – Part 1

4) Session on Open-Source Software – Part 1

Week 11: Data Analysis Process – Part1

1) Session 27: Data Gathering | Data Analysis

- Data Analysis Process
- Import Data from various sources (CSV, excel, JSON, text, SQL)
- Export data in different file formats
- Gather Data through API or Web Scrapping

2) Session 28: Data Assessing and Cleaning

- Data assessing
- Types of unclean data
- Write summary of data
- Types of Assessment
- Manual and Automatic Assessment
- Data Quality Dimension
- Data Cleaning

3) Session on ETL using AWS RDS

- Introduction about Extraction, transform and Load pipeline
- Fetch data from AWS
- Apply transformation on the data
- Upload transformed data into AWS RDS

4) Session on Advanced Web Scrapping using Selenium

- Introduction to Selenium and Chromedriver
- Automated Web scrapping on Smartprix website

Week 12: Data Analysis Process – Part 2

1) Session on Data Cleaning Case Study – Smartphone dataset

- Quality issues
- Tidiness issues
- Data Cleaning

2) Session 29: Exploratory Data Analysis (EDA)

- Introduction to EDA
- Why EDA?

- Steps for EDA
- Univariate Analysis
- Bivariate Analysis
- Feature Engineering

3) Session on Data Cleaning – Part 2

- Data Cleaning on Smartphone Dataset – Continued

4) Session on EDA Case Study – Smartphone Dataset

Week 13: SQL Basics

1) Session 30: Database Fundamentals

- Introduction to Data and Database
- CRUD operations
- Properties of database
- Types of Database
- DBMS
- Keys
- Cardinality of Relationship
- Drawbacks of Database

2) Session 31: SQL DDL Commands

- Xampp Software
- Types of SQL commands
- DDL commands

3) Session on Tableau – Olympics Dataset (Part 1)

- Download and Install Tableau
- Getting Started with Tableau Desktop
- Overview of Olympic datasets
- Create Dashboards using Tableau

Week 14: SQL Continued – Part 1

1) Session 32: SQL DML commands

- MySQL workbench
- INSERT
- SELECT
- UPDATE
- DELETE
- Functions in SQL

2) Session 33: SQL Grouping and Sorting

- Sorting Data
- ORDER BY
- GROUP BY
- GROUP BY on multiple columns
- HAVING clause
- Practice on IPL Dataset

3) Session on Tableau – Part 2

- Tableau Basics
 - i. Importing Data*
 - ii. Measures and Dimensions*
 - iii. Sheet, dashboard, story*
 - iv. Levels of Granularity*
 - v. Different types of charts*
 - vi. Datetime*
- Hierarchical level of granularity
- Common filters
- Calculated fields and Table Calculations
- Working with Geographical columns
- Dashboard and Interactive filters
- Blending and Dual axis chart
- Connecting to a remote database

Week 15: SQL Continued – Part 2

1) Session 34: SQL Joins

- Introduction to SQL joins
- Types of Joins (Cross, inner, left, right, full outer)
- SQL hands-on on joins
- SET operations
- SELF join
- Query execution order
- Practice questions

2) Session on SQL Case Study 1 – Zomato Dataset

- Understanding Dataset through diagram
- Solving SQL Questions on Zomato Dataset

3) Session 35: Subqueries in SQL

- What is Subquery
- Types of Subqueries
- Independent and Correlated subquery

4) Session on Making a Flight Dashboard using Python and SQL

- How to Connect MYSQL through Python
- Run SQL queries with Python
- Creating a dynamic dashboard with Streamlit on Flights dataset

5) Session on SQL Interview Questions – Part 1

- Database Server Vs Database Client
- Database Engines
- Components of DBMS
- What is Collation?
- COUNT(*) vs COUNT(col)
- Dealing with NULL values
- DELETE Vs TRUNCATE
- Anti joins
- Non-equi joins
- Natural joins
- All and Any operators
- Removing Duplicate Rows
- Metadata Queries

Week 16: Advanced SQL

1) Session 36: Window Functions in SQL

- What are Window functions?
- OVER(), RANK(), DENSE_RANK(), ROW_NUMBER(), FIRST_VALUE(), LAST_VALUE()
- Concept of Frames
- LAG(), LEAD()

2) Session 37: Windows Functions Part 2

- Ranking
- Cumulative sum and average
- Running average
- Percent of total

3) Session 37: Windows Functions Part 3

- Percent Change

- Quantiles/Percentiles
- Segmentation
- Cumulative Distribution
- Partition by multiple columns

4) Session on Data Cleaning using SQL | Laptop Dataset

- Basic level Data Cleaning and Data exploration using SQL
- Why to use SQL for Data Cleaning
- String Data types
- Wildcards
- String Functions
- Data Cleaning

5) Session on EDA using SQL | Laptop Dataset

- EDA on numerical and categorical columns
- Plotting
- Categorical – Categorical Analysis
- Numerical – Numerical Analysis

Week 17: Descriptive Statistics

1) Session 38: Descriptive Statistics Part 1

- What is Statistics?
- Types of Statistics
- Population vs Sample
- Types of Data
- Measures of central tendency
- Measure of Dispersion
- Coefficient of variation
- Graphs for Univariate Analysis
- Frequency Distribution table
- Graphs for bivariate Analysis
- Categorical – Categorical Analysis
- Numerical – Numerical Analysis
- Categorical – Numerical Analysis

2) Session on Datetime in SQL

- Remaining topics of EDA using SQL (numerical – categorical, missing values, ppi, price_bracket, one hot encoding)
- Temporal Data types

- Creating and Populating Temporal Tables
- DATETIME Functions
- Datetime Formatting
- Type Conversation
- DATETIME Arithmetic
- TIMESTAMP VS DATETIME
- Case Study – Flights

Week 18: Descriptive Statistics continued

1) Session 39: Descriptive Statistics part 2

- Quantiles and Percentiles
- Five Number Summary
- Boxplots
- Scatterplots
- Covariance
- Correlation
- Correlation vs Causation
- Visualizing multiple variables

2) Session 40: Probability Distribution Functions (PDF, CDF, PMF)

- Random Variables
- Probability Distributions
- Probability Distribution Functions and its types
- Probability Mass Function (PMF)
- Cumulative Distribution Function (CDF) of PMF
- Probability Density Function (PDF)
- Density Estimation
- Parametric and Non-parametric Density Estimation
- Kernel Density Estimate (KDE)
- Cumulative Distribution Function (CDF) of PDF.

3) Session on SQL Datetime Case Study on Flights Dataset

4) Session on Database Design | SQL Data Types | Database Normalization

- Different SQL Datatypes (Numeric, Text, Datetime, Misc)
- Database Normalization
- ER Diagram

Week 19: Probability Distributions

1) Session 41: Normal Distribution

- How to use PDF in Data Science?
- 2D density plots
- Normal Distribution (importance, equation, parameter, intuition)
- Standard Normal Variate (importance, z-table, empirical rule)
- Properties of Normal Distribution
- Skewness
- CDF of Normal Distribution
- Use of Normal Distribution in Data Science

2) Session 42: Non-Gaussian Probability Distributions

- Kurtosis
- Excess Kurtosis and Types of kurtosis
- QQ plot
- Uniform Distribution
- Log-normal distribution
- Pareto Distribution
- Transformations
 - i. *Mathematical Transformation*
 - ii. *Function Transformer*
 - iii. *Log Transform*
 - iv. *Reciprocal Transform / Square or sqrt Transform*
 - v. *Power Transformer*
 - vi. *Box-Cox Transform*
 - vii. *Yeo-Johnson Transformation*

3) Session on views and User Defined Functions in SQL

- What are views?
- Types of views
- User Defined Functions (Syntax, Examples, Benefits)

4) Session on Transactions and Stored Procedures

- Stored Procedures
- Benefits of using stored procedures
- Transactions (Commit, rollback, savepoint)
- ACID properties of a Transaction

Week 20: Inferential Statistics

1) Session 43: Central Limit Theorem

- Bernouli Distribution
- Binomial Distribution
 - i. *PDF formula*
 - ii. *Graph of PDF*
 - iii. *Examples*
 - iv. *Criteria*
 - v. *Application in Data Science*
- Sampling Distribution
- Intuition of Central Limit Theorem (CLT)
- CLT in code
- Case study
- Assumptions of making samples

2) Session on Central Limit Theorem Proof

3) Session 44: Confidence Intervals

- Population vs Sample
- Parameter vs Estimate
- Point Estimate
- Confidence Interval
 - i. *Ways to Calculate CI*
 - ii. *Applications of CI*
 - iii. *Assumptions of z-procedure*
 - iv. *Formula and Intuition of Z-procedure*
 - v. *Interpreting CI*
 - vi. *T-procedure and t-distribution*
 - vii. *Confidence Intervals in code*

Week 21: Hypothesis Testing

1) Session 45: Hypothesis Testing (Part 1)

- Key idea of hypothesis testing
- Null and alternate hypothesis
- Steps in Hypothesis testing
- Performing z-test
- Rejection region and Significance level

- Type-1 error and Type-2 Error
- One tailed vs. two tailed test
- Applications of Hypothesis Testing
- Hypothesis Testing in Machine Learning

2) **Session 46: Hypothesis Testing (Part 2) | p-value and t-tests**

- What is p-value?
- Interpreting p-value
- P-value in the context of z-test
- T-test
- Types of t-test
 - Single sample t-Test*
 - Independent 2-sample t-Test*
 - Paired 2 sample t-test*
 - Code examples of all of the above*

3) **Session on Chi-square test**

- Chi-square distribution (Definition and Properties)
- Chi-square test
- Goodness of fit test (Steps, Assumptions, Examples)
- Test for Independence (Steps, Assumptions, Examples)
- Applications in machine learning

4) **Session on ANOVA**

- Introduction
- F-distribution
- One-way ANOVA
 - Steps*
 - Geometric Intuition*
 - Assumptions*
 - Python Example*
- Post – Hoc test
- Why t-test is not used for more than 3 categories?
- Applications in Machine Learning

Week 22: Linear Algebra

1) **Session on Tensors | Linear Algebra part 1(a)**

- What are tensors?
- 0D, 1D and 2D Tensors

- Nd tensors
- Rank, axes and shape
- Example of 1D, 2D, 3D, 4D, 5D tensors

2) Session on Vectors | Linear Algebra part 1(b)

- What is Linear Algebra?
- What are Vectors?
- Vector example in ML
- Row and Column vector
- Distance from Origin
- Euclidean Distance
- Scalar Addition/Subtraction (Shifting)
- Scalar Multiplication/Division [Scaling]
- Vector Addition/Subtraction
- Dot product
- Angle between 2 vectors
- Equation of a Hyperplane

3) Linear Algebra Part 2 | Matrices (computation)

- What are matrices?
- Types of Matrices
- Matrix Equality
- Scalar Operation
- Matrix Addition, Subtraction, multiplication
- Transpose of a Matrix
- Determinant
- Minor
- Cofactor
- Adjoint
- Inverse of Matrix
- Solving a system of Linear Equations

4) Linear Algebra Part 3 | Matrices (Intuition)

- Basis vector
- Linear Transformations
- Linear Transformation in 3D
- Matrix Multiplication as Composition
- Test of Commutative Law
- Determinant and Inverse

- Transformation for non-square matrix?
- Why only square matrix has inverse?
- Why inverse is possible for non-singular matrices only?

Week 23: Linear Regression

1) Session 48: Introduction to Machine Learning

- About Machine Learning (History and Definition)
- Types of ML
 - i. *Supervised Machine Learning*
 - ii. *Unsupervised Machine Learning*
 - iii. *Semi supervised Machine Learning*
 - iv. *Reinforcement Learning*
- Batch/Offline Machine Learning
- Disadvantages of Batch learning
- Online Machine Learning
 - i. Importance
 - ii. When to use and how to use
 - iii. Learning Rate
 - iv. Out of core learning
 - v. Disadvantages
- Batch vs Online learning
- Instance based learning
- model-based learning
- Instance vs model-based learning
- Challenges in ML
 - i. Data collection
 - ii. Insufficient/Labelled data
 - iii. Non-representative data
 - iv. Poor quality data
 - v. Irrelevant features
 - vi. Overfitting and Underfitting
 - vii. Offline learning
 - viii. Cost
- Machine Learning Development Life-cycle
- Different Job roles in Data Science
- Framing a ML problem | How to plan a Data Science project

2) Session 49: Simple Linear regression

- Introduction and Types of Linear Regression
- Simple Linear Regression
- Intuition of simple linear regression
- Code example
- How to find m and b?
- Simple Linear Regression model code from scratch
- Regression Metrics
 - i. *MAE*
 - ii. *MSE*
 - iii. *RMSE*
 - iv. *R2 score*
 - v. *Adjusted R2 score*

3) Session 50: Multiple Linear Regression

- Introduction to Multiple Linear Regression (MLR)
- Code of MLR
- Mathematical Formulation of MLR
- Error function of MLR
- Minimizing error
- Error function continued
- Code from scratch

4) Session on Optimization the Big Picture

- Mathematical Functions
- Multivariable Functions
- Parameters in a Function
- ML models as Mathematical Function
- Parametric Vs Non-Parametric ML models
- Linear Regression as a Parametric ML model
- Loss Function
- How to select a good Loss Function?
- Calculating Parameters from a Loss Function
- Convex And Non-Convex Loss Functions
- Gradient Descent
- Gradient Descent with multiple Parameters
- Problems faced in Optimization
- Other optimization techniques

5) Session on Differential Calculus

- What is differentiation?
- Derivative of a constant
- Cheatsheet
- Power Rule
- Sum Rule
- Product Rule
- Quotient Rule
- Chain Rule
- Partial Differentiation
- Higher Order Derivatives
- Matrix Differentiation

Week 24: Gradient Descent

1) Session 51: Gradient descent from scratch

- What is Gradient Descent?
- Intuition
- Mathematical Formulation
- Code from scratch
- Visualization 1
- Effect of Learning Rate
- Adding m into the equation
- Effect of Loss function
- Effect of Data

2) Session 52 (part 1): Batch Gradient Descent

- Types of Gradient Descent
- Mathematical formulation
- Code from scratch

3) Session 52 (part 2): Stochastic Gradient Descent

- Problems with Batch GD
- Stochastic GD
- Code from scratch
- Time comparison
- Visualization

- When to use stochastic GD
- Learning schedules
- Sklearn documentation

4) Session 52 (part 3): Mini-batch Gradient Descent

- **Introduction**
- **Code**
- **Visualization**

5) Doubt Clearance session on Linear Regression

Week 25: Regression Analysis

1) Session on Regression Analysis (Part 1)

- What is Regression Analysis?
- Why Regression Analysis is required?
- What's the Statistic connection with Regression analysis?
- Inference vs Prediction
- Statsmodel Linear Regression
- TSS, RSS, and ESS
- Degree of freedom
- F-statistic and Prob(F-statistic)

2) Session on Regression Analysis (Part 2)

- F -test for overall significance
- R-squared (Goodness of fit)
- Adjusted R-squared
- T – Statistic
- Confidence Intervals for Coefficients

3) Session on Polynomial Regression

- Why we need Polynomial Regression?
- Formulation of Polynomial Regression
- Polynomial Regression in python

4) Session on Assumptions of Linear Regression

- Assumptions of Linear Regression
 - i. Linearity
 - ii. Normality of Residuals
 - iii. Homoscedasticity
 - iv. No Autocorrelation
 - v. No or little Multicollinearity

- What happens when these assumptions failed?
- How to check each of these assumptions?
- What to do when an assumption fails?
- Standard Error

5) Session 53: Multicollinearity

- What is multicollinearity?
- When is Multicollinearity bad?
- Multicollinearity (Mathematically)
- Perfect and non-perfect Multicollinearity
- Types of multicollinearity
- How to detect Multicollinearity
- Correlation
- VIF (Variance Inflation Factor)
- Condition Number
- How to remove Multicollinearity

Week 26: Feature Selection

1) Session 54: feature Selection Part 1

- What is Feature Selection?
- Why to do Feature Selection?
- Types of Feature Selection
- Filter based Feature Selection
 - i. Duplicate Features*
 - ii. Variance Threshold*
 - iii. Correlation*
 - iv. ANOVA*
 - v. Chi-Square*
 - vi. Advantages and Disadvantages*

2) Session 55: Feature Selection Part 2

- Wrapper method
- Types of wrapper method
- Exhaustive Feature Selection/Best Subset Selection
- Sequential Backward Selection/Elimination
- Sequential Forward Selection
- Advantages and Disadvantages

3) Session on Feature Selection part 3

- Embedded Methods
 - i. Linear Regression*
 - ii. Tree based models*
 - iii. Regularized Models*
- Recursive Feature Elimination
- Advantages and Disadvantages

Week 27: Regularization

1) Session on Regularization Part 1 | Bias-Variance Tradeoff

- Why we need to study Bias and Variance
- Expected Value and Variance
- Bias and Variance Mathematically

2) Session on Regularization Part 1 | What is Regularization

- Bias Variance Decomposition
- Diagram
- Analogy
- Code Example
- What is Regularization?
- When to use Regularization?

3) Ridge Regression Part 1

- Types of Regularization
- Geometric Intuition
- Sklearn Implementation

4) Ridge Regression Part 2

- Ridge Regression for 2D data
- Ridge Regression for nD data
- Code from scratch

5) Ridge Regression Part 3

- **Ridge regression using Gradient Descent**

6) Ridge Regression Part 4

- 5 Key Understandings
 - i. How do the coefficients get affected?*
 - ii. Higher values are impacted more*
 - iii. Bias variance tradeoff*
 - iv. Contour plot*
 - v. Why is it called Ridge?*

7) Lasso Regression

- Intuition
- Code example
- Lasso regression key points

8) Session on Why Lasso Regression creates Sparsity?

9) ElasticNet Regression

- Intuition
- Code example

10) Doubt Clearance session on regularization

Week 28: K Nearest Neighbors

1) Session on K nearest Neighbors Part 1

- KNN intuition
- Code Example
- How to select K?
- Decision Surface
- Overfitting and Underfitting in KNN
- Limitations of KNN

2) Session on coding K nearest Neighbors from scratch

3) Session on How to draw Decision Boundary for Classification problems

4) Session on Advanced KNN Part 2

- KNN Regressor
- Hyperparameters
- Weighted KNN
- Types of Distances (Euclidean and Manhattan)
- Space and Time Complexity
- KD-Tree

5) Classification Metrics Part 1

- Accuracy
- Accuracy of multi-classification problems
- How much accuracy is good?
- Problem with accuracy
- Confusion matrix
- Type 1 error
- Type 2 error
- Confusion matrix of multi-classification problems

- When accuracy is misleading

6) Classification Metrics Part 2

- Precision
- Recall
- F1 score
- Multi class Precision and Recall
- Multi class F1 score

Week 29: PCA

1) Session on Curse of Dimensionality

2) PCA part 1

- Introduction
- Geometric Intuition of PCA
- Why is Variance important?

3) PCA part 2

- Mathematical Problem formulation
- What is covariance and covariance matrix?
- Matrices as Linear Transformation
- EigenVectors and Eigenvalues
- Step by step solution of PCA
- How to transform points?
- PCA step-by-step code in python

4) PCA Part 3

- Practical example on MNIST dataset
- PCA demo with sklearn
- Visualization
- What is explained variance
- Find optimum number of Principal components
- When PCA does not work?

5) Session on Eigen vectors and Eigen Values

- What are Matrices?
- What are Eigen Vectors and Eigen Values?
- Intuition – Axis of rotation
- How to calculate Eigen Vectors and Eigen Values
- Properties
- Eigen Vectors in PCA

6) Session on Eigen Decomposition and PCA variants

- Types of PCA variants
- What are some special matrices?
 - i. *Diagonal Matrix*
 - ii. *Orthogonal Matrix*
 - iii. *Symmetric Matrix*
- Matrix as Linear Transformation Visualization tool
- Matrix Composition
- Matrix Decomposition
- Eigen decomposition
- Eigen decomposition of Symmetric Matrix (Spectral Decomposition)
- Advantages of Eigen decomposition
- Kernel PCA
- Code example of Kernel PCA

7) Session on eigen Singular Value Decomposition (SVD)

- Intuition of Non-Square Matrix
- Rectangular Diagonal Matrix
- What is SVD
- Applications of SVD
- SVD - The intuition of the mathematical equation
- Relationship with Eigen Decomposition
- Geometric Intuition of SVD
- How to calculate SVD
- SVD in PCA

Week 30: Model Evaluation & Selection

1. ROC Curve in Machine Learning

- a. ROC AUC Curve and its requirements
- b. Confusion matrix
- c. True Positive Rate (TPR)
- d. False Positive Rate (FPR)
- e. Different cases of TPR & FPR

2. Session on Cross Validation

- a. Why do we need Cross Validation?

- b. Hold-out approach
- c. Problem with Hold-out approach
- d. Why is the Hold-out approach used?
- e. Cross Validation
 - i. Leave One Out Cross Validation (LOOCV)
 - 1. Advantages
 - 2. Disadvantages
 - 3. When to use
 - ii. K-Fold Cross Validation
 - 1. Advantages
 - 2. Disadvantages
 - 3. When to use
 - iii. Stratified K-Fold CV

3. Session on Data Leakage

- a. What is it and what is the problem
- b. Ways in which Data Leakage can occur
- c. How to detect
- d. How to remove Data Leakage
- e. Validation set

4. Session on Hyperparameter Tuning

- a. Parameter vs Hyperparameter
- b. Why the word "hyper" in the term
- c. Requirements
 - i. Grid Search CV
 - ii. Randomized Search CV
- d. Can this be improved?

Week 31: Naive Bayse

1. Crash course on Probability Part 1

- a. 5 important terms in Probability
 - i. Random Experiment
 - ii. Trials
 - iii. Outcome
 - iv. Sample Space
 - v. Event
- b. Some examples of these terms

- c. Types of events
- d. What is probability
- e. Empirical vs Theoretical probability
- f. Random variable
- g. Probability distribution of random variable
- h. Mean of 2 random variable
- i. Variance of Random variable

2. Crash course on Probability Part 2

- a. Venn diagrams
- b. Contingency table
- c. Joint probability
- d. Marginal probability
- e. Conditional probability
- f. Intuition of Conditional Probability
- g. Independent vs Dependent vs Mutually Exclusive Events
- h. Bayes Theorem

3. Session 1 on Naive Bayes

- a. Intuition
- b. Mathematical formulation
- c. How Naive Bayes handles numerical data
- d. What if data is not Gaussian
- e. Naive Bayse on Textual data

4. Session 2 on Naive Bayes

- a. What is underflow in computing
- b. Log Probabilities
- c. Laplace Additive Smoothing
- d. Bias Variance Trade off
- e. Types
 - i. Gaussian Naive Bayes
 - ii. Categorical Naive Bayes
 - iii. Multinomial Naive Bayes

5. Session 3 on Naive Bayes

- a. Probability Distribution related to Naive Bayes
 - i. Bernoulli Distribution
 - ii. Binomial Distribution
 - iii. Categorical distribution / Multinoulli distribution

- iv. Multinomial Distribution
 - v. Why do we need these distributions?
 - b. Categorical Naive Bayes
 - c. Bernoulli Naive Bayes
 - d. Multinomial Naive Bayes
 - e. Out of Core Naive Bayes
- 6. End to End Project | Email Spam Classifier**

Week 32 : Logistics Regression

1. Session 1 on Logistic Regression

- a. Introduction
- b. Some Basic Geometry
- c. Classification Problem
 - i. Basic Algorithm
 - ii. Updation in Basic Algorithm
- d. Sigmoid Function
- e. Maximum Likelihood
- f. Log Loss
- g. Gradient Descent
- h. summary

2. Session on Multiclass Classification using Logistic Regression

- a. What is Multiclass Classification
- b. How Logistic Regression handles Multiclass Classification Problems.
- c. One vs Rest (OVR) Approach
 - i. Intuition
 - ii. Code
- d. SoftMax Logistic Regression Approach
 - i. SoftMax Function
 - ii. Code
- e. When to use what?
- f. Tasks

3. Session on Maximum Likelihood Estimation

- a. Recap
- b. Some Examples
 - i. Example 1 – Coin Toss

- ii. Example 2 – Drawing balls from bag
 - iii. Example 3 – Normal Distribution
 - c. Probability Vs Likelihood
 - d. Maximum Likelihood Estimation
 - e. MLE for Normal Distribution
 - f. MLE in Machine Learning
 - g. MLE in Logistic Regression
 - h. Some Important Questions
- 4. Session 3 on Logistic Regression**
- a. Maximum Likelihood in Logistic Regression
 - b. FAQ on MLE (Maximum Likelihood Estimation)
 - i. Is MLE a general concept applicable to all ML algorithms?
 - ii. How is MLE related to the concept of loss functions?
 - iii. Why does the loss function exist, why don't we maximize likelihood?
 - iv. Why study about maximum likelihood at all?
 - c. An interesting task for you
 - d. Assumptions of Logistics Regression
 - e. Odds and Log(Odds)
 - f. Another interpretation of Logistic Regression
 - g. Polynomial Features
 - h. Regularization in Logistic Regression
- 5. Logistic Regression Hyperparameters**

Week 33 : Support Vector Machines (SVM)

- 1. SVM Part 1 – Hard Margin SVM**
- a. Introduction
 - b. Maximum Margin Classifier
 - c. Support Vectors
 - d. Mathematical Formulation
 - e. How to solve this?
 - f. Prediction
 - g. Coding Example
 - h. Problems with Hard Margin SVM

2. SVM Part 2 | Soft Margin SVM

- a. Problems with Hard Margin SVM
- b. Slack Variable
- c. Soft Margin SVM
- d. Introduction of C
- e. Bias-Variance Trade Off
- f. Code Example
- g. Relation with Logistic Regression

3. Session on Constrained Optimization Problem

- a. Problem with SVC
- b. Kernel's Intuition
- c. Coding examples of Kernel
- d. Types of Kernels
- e. Why is it called Trick?
- f. Mathematics of SVM

4. Session on SVM Dual Problem

- a. SVM in n Dimensions
- b. Constrained Optimization Problems
- c. Karush Kuhn Tucker Conditions
- d. Concept of Duality
- e. SVM Dual Problem
- f. Dual Problem Derivation
- g. Observations

5. Session on Maths Behind SVM Kernels

- a. SVM Dual Formulation
- b. The Similarity Perspective
- c. Kernel SVM
- d. Polynomial Kernel
 - i. Trick
 - ii. What about the other Polynomial terms
- e. RBF Kernel
 - i. Local Decision Boundary
 - ii. Effect of Gamma
- f. Relationship Between RBF and Polynomial K...
- g. Custom Kernels
 - i. String kernel

- ii. Chi-square kernel
- iii. Intersection kernel
- iv. Hellinger's kernel
- v. Radial basis function network (RBFN) kernel
- vi. Spectral kernel

Extra Sessions – Feature Engineering

1. Session on Handling Missing Values Part – 1

- a. Feature Engineering
 - i. Feature Transformation
 - ii. Feature Construction
 - iii. Feature Extraction
 - iv. Feature Selection
- b. Types of Missing Values
 - i. Missing Completely at random
 - ii. Missing at Random
 - iii. Missing Not at Random
- c. Techniques for Handling Missing Values
 - i. Removing Missing Values
 - ii. Imputation
- d. Complete Case Analysis

2. Session 2 on Handling Missing Data

- a. Univariate Imputation – Numerical Data
 - i. Mean Imputation
 - ii. Median Imputation
- b. Univariate Imputation – Arbitrary Value & End Distribution Value
- c. Univariate Imputation – Categorical Data
 - i. Mode Imputation
 - ii. Missing Category Imputation
- d. Univariate Imputation – Random (Numerical + Categorical)
- e. Missing Indicator

3. Session 3 on Handling Missing Data – Multivariate Imputation

- a. KNN Imputer
 - i. Steps in KNN Imputation
 - ii. Advantages and Disadvantages in KNN Imputation
- b. Iterative Imputer

- i. MICE
- ii. When To Use
- iii. Advantages and Disadvantages
- iv. Demonstration of MICE algorithm
- v. Steps involved in Iterative Imputer
- vi. Important parameter in Iterative Imputer

Week 34 : Decision Trees

1. Session 1 on Decision Tree

- a. Introduction
- b. Intuition behind DT
- c. Terminology in Decision Tree
- d. The CART Algorithm - Classification
- e. Splitting Categorical Features
- f. Splitting Numerical Features
- g. Understanding Gini Impurity?
- h. Geometric Intuition of DT

2. Session 2 on Decision Tree

- a. CART for Regression
- b. Geometric Intuition of CART
- c. How Prediction is Done
- d. Advantages & Disadvantages of DT
- e. Project Discussion - Real Estate

3. Session 3 on Decision Tree

- a. Feature Importance
- b. The Problem of Overfitting
- c. Why Overfitting happens
- d. Unnecessary nodes
- e. Pruning & its types
 - i. Pre-pruning
 - ii. Post Pruning
- f. Cost Complexity Pruning

4. Session on Decision Tree Visualization

- a. dtreeviz Demo - Coding

Week 35 : Ensemble Methods – Introduction

1. Introduction to Ensemble Learning

- a. Intuition
- b. Types of Ensemble Learning
- c. Why it works?
- d. Benefits of Ensemble
- e. When to use Ensemble

2. Bagging Part 1 – Introduction

- a. Core Idea
- b. Why use Bagging?
- c. When to use Bagging?
- d. Code Demo

3. Bagging Part 2 – Classifier

- a. Intuition through Demo app
- b. Code Demo

4. Bagging Part 3 – Regressor

- a. Core Idea
- b. Intuition through demo web app
- c. Code Demo

5. Random Forest : Session 1

- a. Introduction to Random Forest
- b. Bagging
- c. Random Forest Intuition
- d. Why Random Forest Works?
- e. Bagging vs. Random Forest
- f. Feature Importance

6. Random Forest : Session 2

- a. Why Ensemble Techniques work?
- b. Random Forest Hyperparameters
- c. OOB Score
- d. Extremely Randomized Trees

- e. Advantages and Disadvantages of Random Forest

Week 36 : Gradient Boosting

1. Gradient Boosting : Session 1

- a. Boosting
- b. What is Gradient Boosting
- c. How
- d. What
- e. Why

2. Gradient Boosting : Session 2

- a. How Gradient Boosting works?
- b. Intuition of Gradient Boosting
- c. Function Space vs. Parameter Space
- d. Direction of Loss Minimization
- e. How to update the function
- f. Iterate
- g. Another perspective of Gradient Boosting
- h. Difference between Gradient Boosting and Gradient Descent

3. Gradient Boosting : Session 3 (Classification - 1)

- a. Classification vs. Regression
- b. Prediction

4. Gradient Boosting for Classification - 2 | Geometric Intuition

- a. Geometric Intuition

5. Gradient Boosting for Classification - 3 | Math Formulation

- a. Step 0 : Loss Function
- b. Step 1 : Minimize Loss Function to get $F_0(x)$
- c. Step 2 :
 - i. Pseudo Residuals
 - ii. Training Regression Tree
 - iii. Compute Lambda for all leaf nodes
 - iv. Update the Model
- d. Step 3 : Final Model
- e. Log(odds) vs Probability

Capstone Project

1. Session 1 on Capstone Project | Data Gathering

- a. Project overview in details
 - b. Gather data for the project
 - c. Details of the data
- 2. Session 2 on Capstone Project | Data Cleaning**
 - a. Merging House and Flats Data
 - b. Basic Level Data Cleaning
- 3. Session 3 on Capstone Project | Feature Engineering**
 - a. Feature Engineering on Columns:
 - i. additionalRoom
 - ii. areaWithType
 - iii. agePossession
 - iv. furnishDetails
 - v. features : luxury Score
- 4. Session 4 on Capstone Project | EDA**
 - a. Univariate Analysis
 - b. PandasProfiling
 - c. Multivariate Analysis
- 5. Session 5 on Capstone Project | Outlier Detection and Removal**
 - a. Outlier Detection And Removal
- 6. Session 6 on Capstone Project | Missing Value Imputation**
 - a. Outlier Detection and Removal on area and bedroom
 - b. Missing Value Imputation
- 7. Session 7 on Capstone Project | Feature Selection**
 - a. **Feature Selection**
 - i. Correlation Technique
 - ii. Random Forest Feature Importance
 - iii. Gradient Boosting Feature Importance
 - iv. Permutation Importance
 - v. LASSO
 - vi. Recursive Feature Elimination
 - vii. Linear Regression with Weights
 - viii. SHAP (Explainable AI)
 - b. **Linear Regression – Base Model**
 - i. One-Hot Encoding
 - ii. Transformation
 - iii. Pipeline for Linear Regression

c. SVR

8. Session 8 on Capstone Project | Model Selection & Productionalization

a. Price Prediction Pipeline

- i. Encoding Selection
 1. Ordinal Encoding
 2. OHE
 3. OHE with PCA
 4. Target Encoding

ii. Model Selection

b. Price Prediction Web Interface –Streamlit

9. Session 9 on Capstone Project | Building the Analytics Module

- a. geo map
- b. word cloud amenities
- c. scatterplot -> area vs price
- d. pie chart bhk filter by sector
- e. side by side boxplot bedroom price
- f. distplot of price of flat and house

10. Session 10 on Capstone Project | Building the Recommender System

- a. Recommender System using TopFacilities
- b. Recommender System using Price Details
- c. Recommender System using LocationAdvantages

11. Session 11 on Capstone Project | Building the Recommender System Part 2

- a. Evaluating Recommendation Results
- b. Web Interface for Recommendation (Streamlit)

12. Session 12 on Capstone Project | Building the Insights Module

13. Session 13 on Capstone Project | Deploying the application on AWS

XGBoost (Extreme Gradient Boosting)

1. Introduction to XGBoost

- Introduction
- Features
 - Performance
 - Speed

- Flexibility

2. XGBoost for Regression

- a. Regression Problem Statement
- b. Step-by-Step Mathematical Calculation

3. XGBoost for Classification

- a. Classification Problem Statement
- b. Step-by-Step Mathematical Calculation

4. The Complete Maths of XGBoost

- a. Prerequisite & Disclaimer
- b. Boosting as an Additive Model
- c. XGBoost Loss Function
- d. Deriving Objective Function
- e. Problem With Objective Function and Solution
 - i. The Taylor series
 - ii. Applying Taylor Series
 - iii. Simplification
- f. Output Value for Regression
- g. Output Value for Classification
- h. Derivation of Similarity Score
- i. Final Calculation of Similarity Score

----- DSMP 1.0 Ends -----

MLOps Curriculum (By Pranjal Sir)

Week 1: Introduction to MLOps and ML-DLC

Introduction to MLOps: Understanding what is MLOps and why is it an important field

Maintainable Code Development: Understanding version control and using tools like Git

Challenges in ML Model Deployment: Overview of ML life cycle stages, challenges in model deployment, approaches to deploying ML models

MLOps Best Practices: Industry standards and guidelines (high level overview of the next 8 weeks)

1. Session 1: Introduction to MLOps
 - a. Reality of AI in the market
 - b. Introduction
 - i. Standard ML Cycle
 - ii. What is DevOps?
 - iii. What is MLOPs?
 - c. Machine Learning Lifecycle
 - d. Introduction to Version Control
 - i. Key aspects of Version Control
 - ii. Types of Version Control Systems
 - e. Next two weeks plan
2. Session 2: Version Control
 - a. Using GitHub for Version Control
 - i. What is GitHub?
 - ii. Setting Up GitHub
 - iii. Creating a Repository
 - iv. Cloning a Repository
 - v. Making Changes
 - vi. Committing Changes
 - vii. Pushing Changes
 - viii. Branching
 - ix. Pull Requests
 - x. Collaborating with Others
 - b. Revisiting ML Cycle
 - i. ML Pipeline Example
 - c. Industry Trivia
3. Doubt Clearance Session 1
 - a. Create an Account on GitHub
 - b. GitHub using GUI: VsCode, Github Desktop
 - c. Assignment on GitHub Fundamentals
 - d. Git Push using CLI
 - e. Solving Error: Head is not in Sync
 - f. 55:10: Touch command

g.

Week 2: ML Reproducibility, Versioning, and Packaging

ML Reproducibility: Ensuring consistent results in ML experiments.

Model Versioning: Tools like MLflow.

Packaging and dependency management: Developing deployable ML packages with dependency management. Example – DS-cookie-cutter

Data Versioning and Management: Tools like DVC (Data Version Control). [Optional]

4. Session 3: Reproducibility

- a. Story
- b. Industry Tools
- c. Cookiecutter
 - i. Step 1: Install the Cookiecutter Library and start a project
 - ii. Step 2: Explore the Template Structure
 - iii. Step 3: Customize the Cookiecutter Variables
 - iv. Step 4: Benefits of Using Cookiecutter Templates in Data Science

5. Session 4: Data Versioning Control

- a. Introduction
- b. Prerequisites
- c. Setup
 - i. Step 1: Initialize a Git repository
 - ii. Step 2: Set up DVC in your project
 - iii. Step 3: Add a dataset to your project
 - iv. Step 4: Commit changes to Git
 - v. Step 5: Create and version your machine learning pipeline
 - vi. Step 6: Track changes and reproduce experiments

6. Doubt Clearance Session 2

- a. Assignment Solution on DVC: 10:19
- b. Doubt Clearance
 - i. DVC with G-Drive 42:50

- ii. DVC Setup Error: 48:45
- iii. Containerization with Virtual Environment 49:40
- iv. Create Version and ML Pipeline: 56:50
- v. DVC Checkout 57:50
- vi. How to which ID(commit) to go to – through commit messages? 1:00:00
- vii. What is Kubernetes?
- viii. Not able to understand by reading documentation 1:04:30
- ix. Getting no of commits 11k+ 1:09:40

Week 3: End-to-end ML lifecycle management

Setting up MLFlow: Understand MLFlow and its alternatives in depth.

Life cycle components: Projects, model registry, performance tracking.

Best Practices for ML Lifecycle

- 7. Session 5 – ML Pipelines and Experimentation Tracking
 - a. Doubts
 - i. DVC Track by Add
 - ii. Git clone with SSH vs HTTPS
 - b. Recap
 - c. Pipelines + DVC + Experimentation Tracking
 - d. MLFlow
- 8. Session 6 on MLOPs
 - a. Recap of Pipelines – Credit Card Example
 - b. Writing dvc.yml File
 - c. Reproducibility after Data Changes
 - d. Reproducibility after Params Changes
 - e. ML end-2-end Pipeline
 - f. Tools for different Stages of Pipeline
- 9. Doubt Clearance Session 3
 - a. Assignment Solution
 - b. File not found error – Joblib/Data
 - c. Models Not Found error
 - d. Get familiar With Terminal – DVC –help

- e. DVC Repro vs DVC Exp

Week 4: Containerisation and Deployment Strategies

Containerization: Introduce docker internals and usage.

Distributed infrastructure: Introduce Kubernetes and basic internal components.

Online vs. Offline Model Deployment: Kubernetes for online, and batch processing for offline. A/B testing.

Canary Deployment and Blue/Green Strategies: Kubernetes, AWS CodeDeploy.
[Optional]

- 10. Session 7 Continuous Integration
 - a. Philosophy Behind CI/CD
 - b. Setting UP Github Actions for CI/CD
 - i. Workflow setup
 - ii. Integrating CML for Version Control
 - iii. Update settings in GitHub Actions
 - 1. Setting Secret Tokens
- 11. Session 8 Containerisation – Docker
 - a. Containerization
 - b. Docker

Week 5: DAGs in MLOps

Understanding DAGs in MLOps: Dependency management in ML pipelines.

Building and Managing DAGs: Apache Airflow, Kubeflow Pipelines.

Continuous Integration/ Development: Discuss tools like Github Actions

- 12. Session 9 – Continuous Deployment
 - a. Recap of Continuous Integration
 - b. Continuous Delivery/ Deployment
 - i. Credit Card Project Code Example
 - ii. FastApi, pydantic, joblib, uvicorn
 - iii. How to write app.py

- c. Multi Container
- 13. Doubt Clearance Session 4
 - a. Assignments Solution

Week 6: Monitoring, Alerting, Retraining, and Rollback

Continuous Monitoring in MLOps: Prometheus, Grafana.

Alerting Systems

Automated Retraining Strategies: Kubeflow Pipelines

Rollback Design Patterns in MLOps: Feature flags, canary releases.

- 14. Session 10 - Introduction to AWS
 - a. Introduction to AWS Machine Learning and MLOps Services
 - i. Prerequisite
 - ii. IAM
 - b. AWS Sagemaker
 - c. Amazon S3
 - d. AWS Lambda
 - e. Amazon ECR and ECS
 - f. AWS CodePipeline and AWS CodeBuild
 - g. Components for SageMaker
- 15. Session 11 - Deployment on AWS
 - a. Code Demo with Credit Card Project
 - b. Project Code explained
 - i. CI.yml file
 - 1. Configuring AWS Credentials on GitHub secrets
 - 2. Creating ECR Repository Getting URI
 - 3. AWS Region
 - ii. Pulling ECR repo to EC2 machine
 - iii. How to create Workflow YML files
 - iv. Connecting to EC2
 - c. Self Runner
 - d. AWS Actions

Week 7: Scaling and Efficiency in MLOps

AutoML and Hyperparameter Tuning: AutoML tools (e.g., Google AutoML, Sagemaker Autopilot and Azure AML).

Data Consistency and availability: How offline development becomes a online nightmare

Resource Management and Cost Optimization: Discuss AWS Elastic Kubernetes Service (EKS) Autoscaling, Azure Kubernetes Service (AKS) Autoscale.

16. Session 12 – Distributed Infrastructure

- a. Understanding Distributed Computing
- b. Node, Communication, and Concurrency
- c. Why Distributed Computing?
- d. Docker and Microservices
 - i. Fundamentals of Microservices
 - ii. Data Bricks Architecture as Microservice
 - iii. Uber Microservices Architecture

17. Session 13 on – Kubernetes Internals

- a. What is Kubernetes?
- b. Need of Kubernetes: Container orchestration system
- c. Key Concepts
 - i. Pods
 - ii. Nodes
 - iii. Cluster
 - iv. Control Panel
 - 1. API Server
 - 2. Resource Manager
 - 3. Database
 - v. Deployment
 - 1. Deploying with Kubectl

18. MLOps Doubt Clearance Session 6

- a.

Week 8: Final Project

Final Project: Implementing a Full MLOps Pipeline for a Real-world Use Case **Locally**

19. Session 14 – Deployment on Kubernetes

- a. Deployment with Kubectl
 - i. Explained Deployment yaml file
 - ii. Deployment Demo
- b. Deployment Strategies
- c. Service and Load Balancing

20. Session 15 – Seldon Deployments

- a. Introduction to Sheldon
- b. Key Features of Sheldon
- c. Sheldon vs Competitors
- d. Prerequisite: Kubectl and helm
- e. Deployment
 - i. Step 1: Install Seldon Core using Helm
 - ii. Step 2: Define a Simple Machine Learning Model
 - iii. Step 3: Push your model to S3 or Google store
 - iv. Step 4: Define a Seldon Deployment
 - v. Step 5: Deploy the Seldon Deployment
- f. Seldon vs EKS vs K8s
- g. Kubeflow Pipelines
 - i. KubeFlow Pipeline Overview with MNIST Dataset
- h. Apache Airflow

21. MLOps Doubt Clearance Session 7

- a. TBD

Week 9: ML Technical Debt

Understanding and Managing ML Technical Debt: Identifying and addressing technical debt in ML projects.

- 22. Session 16 – Monitoring & Alerting
 - a. TBD
- 23. Session 17 – Rollout & Rollback Strategies
 - a. TBD
- 24. Session on MLOps Interview Questions
- 25. Session 18 – ML Technical Debt
 - a. TBD
- 26. MLOps Doubt Clearance Session 8

Unsupervised Learning

1. KMeans Clustering

- a. Session 1 on KMeans Clustering**
 - i. Plan of Attack (Getting Started with Clustering)
 - ii. Types of ML Learning
 - iii. Applications of Clustering
 - iv. Geometric Intuition of K-Means
 - v. Elbow Method for Deciding Number of Clusters
 - 1. Code Example
 - 2. Limitation of Elbow Method
 - vi. Assumptions of KMeans
 - vii. Limitations of K Means
- b. Session 2 on KMeans Clustering**
 - i. Recap of Last class
 - ii. Assignment Solution
 - iii. Silhouette Score
 - iv. Kmeans Hyperparameters
 - 1. Number of Clusters(k)
 - 2. Initialization Method (K Means++)
 - 3. Number of Initialization Runs (n_init)
 - 4. Maximum Number of Iterations (max_iter)
 - 5. Tolerance (tol)
 - 6. Algorithm (auto, full, ..)

7. Random State

v. K Means ++

c. Session 3 on KMeans Clustering

- i. K-Means Mathematical Formulation (Lloyd's Algorithm)
- ii. K-Means Time and Space Complexity
- iii. Mini Batch K Means
- iv. Types of Clustering
 1. Partitional Clustering
 2. Hierarchical Clustering
 3. Density Based Clustering
 4. Distribution/Model-based Clustering

d. K-Means Clustering Algorithms from Scratch in Python

- i. Algorithms implementation from Scratch in Python

2. Other Clustering Algorithms

a. Session on DBSCAN

- i. Why DBSCAN?
- ii. What is Density Based Clustering
- iii. MinPts & Epsilon
- iv. Core Points, Border Points & Noise Points
- v. Density Connected Points
- vi. DBSCAN Algorithm
- vii. Code
- viii. Limitations
- ix. Visualization

b. Session on Hierarchical Clustering

- i. Need of Other Clustering Methods
- ii. Introduction
- iii. Algorithm
- iv. Types of Agglomerative Clustering
 1. Min (Single-link)
 2. Max (Complete Link)
 3. Average
 4. Ward
- v. How to find the ideal number of clusters

- vi. Hyperparameter
- vii. Code Example
- viii. Benefits/Limitations

c. Session – 1 on Gaussian Mixture Models (GMM)

- i. The Why?
- ii. The What?
- iii. Geometric Intuition
- iv. Multivariate Normal Distribution
- v. Geometric Intuition 2D
- vi. EM(Expectation Minimization) Algorithm
- vii. Python Code

d. Session – 2 on Gaussian Mixture Models

- i. Recap of Session 1
- ii. Covariance Types: Spherical, Diagonal, Full, and Tied
- iii. How to decide n_components?
 - 1. Akaike Information Criterion (AIC)
 - 2. Bayesian Information Criterion(BIC)
 - 3. Likelihood Formula for GMM
 - 4. Python Implementation.
 - 5. Why not Silhouette Score?
- iv. Visualization
- v. Assumptions
- vi. Advantages & Disadvantages
- vii. K Means vs GMM
- viii. DBSCAN vs GMM
- ix. Applications of GMM

e. Session on T-SNE

- i. What is T-SNE?
- ii. Why learn T-SNE?
- iii. Geometric Intuition
- iv. Mathematical Formulation
- v. Code Implementation

2. Session 2 on T-SNE

- i. Mathematical Formulation

- ii. Some Questions!
 - 1. Why use probabilities instead of distances to calculate similarity?
 - 2. Why use Gaussian distribution to calculate similarity in high dimensions?
 - 3. How is variance calculated for each Gaussian distribution?
 - 4. Why use T-distribution in lower dimensions?
- iii. Code Example
- iv. Hyperparameters
 - 1. Perplexity
 - 2. Learning Rate
 - 3. Number of Iterations
- v. Points of Wisdom
- vi. Advantages & Disadvantages

- 1. LDA (Linear Discriminant Analysis)
 - a. **Introduction:** Supervised dimensionality reduction
 - b. **Algorithm Explanation:** Maximizing between-class variance
 - c. **Applications:** Use cases in classification and visualization
 - d. **Comparison:** Differences and similarities with PCA
- 2. Apriori
 - a. **Introduction:** Principles of association rule mining
 - b. **Key Concepts:** Support, Confidence, Lift
 - c. **Algorithm Steps:** Candidate generation, Pruning
 - d. **Applications:** Market Basket Analysis, Recommender Systems

Feature Engineering

1. Session on Encoding Categorical Features - 1

- a. Feature Engineering Roadmap
- b. What is Feature Encoding
- c. Ordinal Encoding
 - i. Code examples in Python
 - ii. Handling Rare Categories
- d. Label Encoding
 - i. Code Example using Sklearn LabelEncoder
- e. One Hot Encoding
 - i. Code Examples using Sklearn OneHotEncoder
 - ii. Handling unknown Category
- f. LabelBinarizer

2. Session on Sklearn ColumnTransformer & Pipeline

- a. What is ColumnTransformer
- b. Code implementation of ColumnTransformer
 - i. OHE
 - ii. Ordinal
- c. SKLearn Pipelines
 - i. Implementing multiple transformations in Pipeline
 - 1. Missing value imputation
 - 2. Encoding Categorical Variables
 - a. Handling rare Categories
 - 3. Scaling
 - 4. Feature Selection
 - 5. Model building
 - 6. Prediction

3. Session on Sklearn Deep Dive

- a. Estimators
- b. Custom Estimators
- c. Mixins
- d. Transformers
- e. Custom Transformer

- f. Composite Transformers
- g. Column transformer
- h. Feature Union
- i. Pipeline

4. Session 2 on Encoding Categorical Features

- a. Count and Frequency Encodes
 - i. CountEncoder Library
- b. Binary Encoder
- c. Target Encoder

5. Session 1 on Discretization

- a. Remaining topics of the last session
 - i. Weight of Evidence
 - ii. Advice on when to use which Encoder
- b. What is Discretization?
- c. Why learn Discretization?
 - i. Reducing Overfitting
 - ii. Handling Non-Linear Relationships
 - iii. Handling Outliers
 - iv. Better Interpretability
 - v. Model Compatibility
- d. Disadvantages of Discretization

6. Session 2 on Discretization

- a. Types of Discretization
 - i. Uniform Binning
 - ii. Quantile Binning
 - iii. K-Means Binning
 - iv. Decision Tree Based Binning
 - v. Custom Binning
 - vi. Threshold Binning (Binarization)

7. Session 1 on Handling Missing Data

- a. Missing Values
- b. The missingo library

- c. Why missing values occur?
- d. Types of missing values
- e. How missing values impact ML models?
- f. How to handle missing values?
 - i. Removing
 - ii. Imputing
- g. Removing Missing Data

8. Session 2 on Handling Missing Data

- a. Removing Missing Values
- b. Missing Indicator
 - i. When to use?
 - ii. When to not use?
- c. Simple Imputer -
 - i. Mean & Median
 - ii. Most Frequent
 - iii. Constant
- d. How to select the best

9. Session 3 on Handling Missing Data - Multivariate Imputer

- a. NaN Euclidean distance
- b. KNN Imputer
 - i. Code and Hyperparameter
 - ii. When to use and not to use.
 - iii. Advantages and Disadvantages
- c. Iterative Imputer
 - i. Code sed Parameters
 - ii. When to use and not to use.
 - iii. Advantages and Disadvantages
- d. Coding Framework to compare different techniques

10. Session on Feature Scaling

- a. What is Feature Scaling
 - i. Why do we need feature scaling?
 - ii. Which algorithms are affected if the features are not scaled?
 - iii. Which algorithms are not affected?
- b. Standardization

11. Session 2 on Feature Scaling

- a. Minmax Scaling
- b. Standardization vs Minmax Scaling
- c. Robust Scaler
- d. Max Absolute Scaler
- e. L2/L1 Normalization
- f. Comparison
- g.

12. Session 1 on Outlier Detection

- a. What are Outliers
- b. Types of Outliers
- c. Impact of Outliers
- d. How to deal with outliers
- e. Outlier Detection Techniques
 - i. Univariate
 - 1. Z-Score
 - 2. IQR and BoxPlot
 - 3. Problem with Univariate Techniques
 - ii. Multivariate Outlier Detection
 - 1. Isolation Forest

13. Session 2 on Outlier Detection

- a. Isolation Forest
- b. Calculation of Anomaly Score
- c. Outlier Detection using KNN
- d. Local vs Global Outliers
- e. Local Outlier Factor(LOF)

14. Session 3 on Outlier Detection

- a. Local Outlier Factor(LOF)
- b. DBSCAN - Code visualization
- c. How do you assess the accuracy?
- d. When should you use which algorithm for outlier detection?

15. Session on Feature Transformation

- a. Why do we need transformations?

- b. What are Feature Transformations
 - i. Problems after transformation
- c. Log transformation
 - i. Algorithms Benefitted
 - ii. When to use
 - iii. When not to use
- d. Square Root Transformation
- e. Reciprocal Transformation
- f. Case Study on Boston Housing Price
- g. Square Transformation
- h. Box-Cox Transform
- i. Yeo Johnson Transform

Advanced XGBoost

Session on Revisiting XGBoost

- 1. Supervised ML
- 2. Stagewise Additive Modelling
- 3. XGBoost Objective Function

Session on XGBoost Regularization

- 1. Recap
- 2. Ways to reduce overfitting in XGBoost
- 3. Parameters
 - a. Gamma
 - b. Max Depth
 - c. Num Estimators
 - d. Early Stopping
 - e. Shrinkage

Session on XGBoost Regularization (Cont.)

- 1. Parameters
 - a. Min Child Weight
 - b. Lambda (and Alpha)
 - c. Subsample
 - d. Col Subsample

Session on XGBoost Optimizations

- 1. Recap
- 2. Reason for slowness in Boosting Algorithms
- 3. Exact Greedy Split Finding

4. Approximate Method For Split Finding
5. Improvements for Optimization
 - a. Reduces No. of Splits
 - b. Calculation of G and H
 - c. Parallel Processing
 - d. Cache Storage
 - e. Quantiles Sketch
 - f. Weighted Quantiles Sketch

How XGBoost Handles Missing Values

1. Handling Missing Values

Competitive Data Science

1. Adaboost

- a. **Introduction:** Overview and intuition of the algorithm
- b. **Components:** Weak Learners, Weights, Final Model
- c. **Hyperparameters:** Learning Rate, Number of Estimators
- d. **Applications:** Use Cases in Classification and Regression

2. Stacking

- a. **Introduction:** Concept of model ensembling
- b. **Steps:** Base Models, Meta-Model, Final Prediction
- c. **Variations:** Different approaches and modifications
- d. **Best Practices:** Tips for effective stacking

3. LightGBM

Session 1 on Introduction to LightGBM

- a. Introduction and core features
- b. Boosting and Objective Function
- c. Histogram-Based Split finding
- d. Best-fit Tree (Leaf-wise growth strategy)
- e. Gradient-based One side sampling(GOSS)
- f. Exclusive Feature Bundling (EFB)

Session 2 on LightGBM (GOSS & EFB)

- a. Recap - Features and Technical Aspects
- b. Revisiting GOSS
- c. EFB

4. CatBoost

Session 1 on CatBoost - Practical Introduction

- a. Introduction
 - b. Advantages and Technical Aspects
 - c. Practical Implementation of CatBoost on Medical Cost Dataset
5. Advanced Hyperparameter Tuning
- a. **Strategies:** Bayesian Optimization
 - b. **Libraries:** Optuna, Hyperopt
 - c. **Practical Tips:** Efficient tuning, Avoiding overfitting
 - d. **Evaluation:** Ensuring robust model performance
6. Participating in a real Kaggle Competition
- a. **Getting Started:** Understanding the problem, Exploring datasets
 - b. **Strategy:** Model selection, Preprocessing, Validation
 - c. **Collaboration:** Teamwork, Sharing, and Learning
 - d. **Submission and Evaluation:** Making effective submissions, Learning from feedback

Miscellaneous Topics

1. NoSQL

- a. **Introduction:** Overview of NoSQL databases
- b. **Types:** Document, Key-Value, Column-Family, Graph
- c. **Use Cases:** When to use NoSQL over SQL databases
- d. **Popular Databases:** MongoDB, Cassandra, Redis, Neo4j

2. Model Explainability

- a. **Introduction:** Importance of interpretable models
- b. **Techniques:** LIME, SHAP, Feature Importance

- c. **Application:** Applying techniques to various models
 - d. **Best Practices:** Ensuring reliable and accurate explanations
- 3. FastAPI
 - a. **Introduction:** Modern, fast web framework for building APIs
 - b. **Features:** Type checking, Automatic validation, Documentation
 - c. **Building APIs:** Steps and best practices
 - d. **Deployment:** Hosting and scaling FastAPI applications
- 4. AWS Sagemaker
 - a. **Introduction:** Fully managed service for machine learning
 - b. **Features:** Model building, Training, Deployment
 - c. **Usage:** Workflow from data preprocessing to model deployment
 - d. **Best Practices:** Optimizing costs and performance
- 5. Handling Imbalanced Data

Session 1 on Imbalanced Data - Introduction

- 1. What is Imbalanced Data
- 2. Problem with Imbalanced Data
- 3. A Quick Detour!
- 4. Why algorithms perform badly on Imbalanced Data?
- 5. Why is imbalanced data a huge deal?
- 6. Techniques
 - a. Under Sampling
 - b. Over Sampling
 - c. SMOTE
 - d. Hybrid Sampling
 - e. Class Weighting
 - f. Cost-Sensitive Learning
 - g. Ensemble Methods
 - h. Threshold Tuning

Session 2 on Imbalanced Data - Oversampling Techniques

- 1. Recap
- 2. Random Oversampling
- 3. SMOTE
- 4. Borderline SMOTE
- 5. ADASYN

Session 2 on Imbalanced Data - Oversampling Techniques Continued

1. SVM SMOTE
2. SMOTENC
3. When to use which technique.

Session 3 on Imbalanced Data - Undersampling Techniques

1. Recap
2. Random Undersampling Technique
3. Tomek Links
 - a. What is Tomek Link
 - b. How does it work
 - c. Code
4. Edited Nearest Neighbors
5. Neighborhood Cleaning Rule
6. Cluster Centroids

Regular Expression - RegEx

Session 1 on Regular Expressions

1. Introduction - What, Why
2. What is a raw string
3. re Module
4. Searching Functions
 - a. match
 - b. search
 - c. findall
 - d. finditer
5. Meta Characters - dot(.), caret(^), dollar (\$)
6. Character Sets
 - a. Custom Character Sets
 - b. Character Ranges
 - c. Character Negation
 - d. Pre-defined Character Sets
7. Quantifiers
 - a. Greedy
 - b. Non Greedy or Lazy

Session 1 on Regular Expressions

1. Grouping
 - a. Captured Groups
 - b. Named-Captured Groups
 - c. Back References
 - d. Non-capture Groups
 - e. Alternation
2. Modifications
 - a. Split

- b. Substitution
 - c.
- 3. Assertions
 - a. Look Ahead
 - b. Look Behind
- 4. Flags

Interview Questions (Sessions)

Statistics

Session 1 on Interview Questions on Statistics

- 1. TBD

Session 2 on Interview Questions on Statistics

- 1. TBD

Session 3 on Interview Questions on Statistics

- 1. TBD

Session 4 on Interview Questions on Statistics

- 1. TBD

Project Based

Session on Project Based Interview Questions

- 1. TBD

Machine Learning

Session 1 on ML Interview Questions

- 1. TBD

Session 2 on ML Interview Questions

- 1. TBD

Session 3 on ML Interview Questions

- 1. TBD

Session 4 on ML Interview Questions

- 1. TBD

SQL

Session 1 on SQL Interview Questions

- 1. TBD

Session 2 on SQL Interview Questions

- 1. TBD

Python

Session 1 on Python Interview Questions

- 1. TBD

MLOPs Revisited (By Nitish Sir)

1. Session 1 MLOps Revisited - Introduction to MLOps

- a. What is MLOPs
- b. Problems in Old Methods of ML Project
 - i. Cricket Score Forecast Project Explanation
 - ii. The Data Problem
 - iii. The code problem
 - iv. The versioning problem
 - v. Automation
 - vi. The Experimentation Problem
 - vii. The Deployment Problem
 - viii. The Drift problem
 - ix. The Infrastructure Problem
 - x. The Collaboration Problem
 - xi. The Legal Problem
- c. Aspects of MLOPs

2. Session 2 on MLOps Revisited - MLOps Tools Stack

- a. Plan and Recap of Core Aspects
- b. Benefits of MLOPs
- c. Challenges in MLOPs
- d. Prerequisites of MLOPs
- e. MLOPs Tool Stack
 - i. End to End MLOps Platforms - AWS Sagemaker, Azure ML ..
 - ii. Data Management Tools
 - 1. Data Ingestion - Kafka
 - 2. Data Transformation - Apache Spark
 - 3. Data Storage

3. Session 3 on MLOps Revisited - MLOps Tools Stack Part 2

- a. Continuing With Data Management Tools
 - i. Data Pipeline - Integration (Airflow)
 - ii. Data Validation
 - iii. Data Versioning - DVC
 - iv. Data Annotation
- b. Data Observability
- c. Data Security
- d. Data Governance
- e. Metadata Store
- f. Feature Store

4. Session 4 on MLOps Revisited - MLOps Tools Stack Part 3

- a. Recap
- b. Code Management
- c. Model Building

- d. AutoML
 - e. Hyperparameter Tuning
 - f. Model Explainability
 - g. Experiment Tracking
 - h. Model Registry
- 5. Session 5 on MLOps Revisited - DevOps Fundamentals**
- a. Revisiting MLOPs
 - b. What are Virtual Machines
 - c. Containerization
 - d. Deploying a container
 - e. Container Orchestration
- 6. Session 6 on MLOps Revisited - DevOps Contd.**
- a. Recap
 - b. Monolithic Architecture
 - c. Microservices
 - d. Kubernetes
 - e. Container Orchestration Services
- 7. Session 7 on MLOps Revisited - MLOps Tool Stack - CI/CD**
- a. Continuous Integration - GitHub Action
 - i. Steps executed during the CI workflow
 - ii. YAML File Explanation
 - b. Continuous Deployment
 - i. Continuous Deployment Workflow
 - ii. Sample Workflow - yaml
 - c. Continuous Deployment vs Continuous Delivery
 - d. Model Serving
- 8. Session 8 on MLOps Revisited - MLOps Tool Stack - Cloud Infrastructure**
- a. Cloud Platform - AWS vs GCP vs Azure
 - b. Provisioning
 - i. Provision Resources
 - 1. Example Terraform Script
 - ii. Configure Environments
 - 1. Example Bash Script
 - iii. Deploy Services
 - iv. Set Up Networking
 - v. Implement Security Measures
 - c. Load Balancers
 - i. What, Why and How
 - ii. Load Balancing Algorithms
 - d. Auto Scaling
- 9. Session 9 on MLOps Revisited - ML Pipelines using DVC**
- a. How to Build an ML Pipeline - Tweet Emotion project
 - b. Working of DVC - dvc yaml file
 - c. Pipeline from scratch

- i. Data ingestion
- ii. Data Preprocessing
- iii. Feature Engineering
- iv. Model Building
- v. Model Evaluation

10. Session 10 on MLOps Revisited - Improving ML Pipelines

- a. Creating Params yaml
- b. Improving Using
 - i. Function
 - ii. Type Hinting
 - iii. Exception Handling
 - iv. Logging

11. Session 11 on MLOps Revisited - Version Control

- a. Steps For Starting a ML Project with MLOPs
- b. Code Versioning - Git
- c. Data Versioning
 - i. Why - Data Versioning?
 - ii. Why Not Git for Data Versioning

12. Session 12 on MLOps Revisited - Data Versioning using DVC

- a. DVC Working
- b. Step-by-Step Process for Data Versioning
 - i. Data Ingestion Project - eCommerce

13. Session 13 on MLOps Revisited - Pipeline x Data Versioning x AWS S3

- a. DVC Pipeline - Data Versioning
 - i. Emotion Detection Project (Cont.)
 - ii. DVC Repro
- b. DVC with AWS S3 storage
 - i. Step by step setup

14. Session 14 (Part 1) on MLOps Revisited - Experiment Tracking using DVC (without Pipeline)

- a. Journey Till Now
- b. Need of experiment Tracking
- c. What is Experiment Tracking
 - i. Benefits
 - ii. Tools for Exp. Tracking
- d. DVC Experiment Tracking Demo (Without Pipeline)

15. Session 14 (Part 2) on MLOps Revisited - Experiment Tracking using DVC (with Pipeline)

- a. DVC Exp. Tracking Demo with Pipeline
 - i. Step by Step Process

- b. DVC repro vs DVC exp run
- 16. Session 15 on MLOps Revisited - Introduction to MLflow**
 - a. Recap
 - b. Limitation of DVC
 - c. MLFlow
 - i. Introduction to MLflow
 - ii. MLFlow demo
 - iii. What can MLflow track
 - iv. MLflow tracking Server
- 17. Session 16 (Part 1) on MLOps Revisited - MLOps Remote Tracking using Dagshub**
 - a. Recap
 - b. Plan of Attack for MLFlow Tracking Remote Server
 - c. Dagsub
 - i. Introduction
 - ii. Demo
- 18. Session 16 (Part 2) on MLOps Revisited - MLflow Remote Tracking using AWS**
 - a. AWS Setup for MLflow
 - i. Create IAM User
 - ii. Create S3 bucket
 - iii. Create EC2 instance
 - b. Steps to start MLFlow server
 - c. Working Demo of MLFlow on AWS
- 19. Session 17 (Part 1) on MLOps Revisited - Autologging in MLflow**
 - a. TBD
- 20. Session 17 (Part 2) on MLOps Revisited - Hyperparameter Tuning in MLflow**
 - a. TBD
- 21. Session 18 on MLOps Revisited - Model Registry**
 - a. TBD
- 22. Session 19 on MLOps Revisited - Mini Project**
 - a. TBD
- 23. Session 20 on MLOps Revisited - Model Serving**
 - a. TBD
- 24. Session 21 on MLOps Revisited - Fundamentals of CI**
 - a. TBD

Note: The schedule is tentative and topics can be added/removed from it in the future.

